

Juliana Cunha

Ph.D. Candidate | Mathematics Department | University of Aveiro

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Research Interests: Paraconsistent Logic · Formal Methods · Algebraic Semantics · Twist Structures

Education

Doctorate in Mathematics · University of Aveiro

Set. 2023 - Present

- Dissertation: “Parametric Paraconsistent Systems and Their Logics”

- Advisors: Alexandre Madeira & Luís S. Barbosa

Master in Mathematics and Applications · University of Aveiro

Set. 2020 - Feb. 2023

- Computing and Informatics Minor

- Thesis: “Towards a Specification Theory for Paraconsistent Transition Systems”

- Advisors: Alexandre Madeira & Luís S. Barbosa

Bachelor in Mathematics · Faculty of Sciences - University of Porto

Set. 2017 - Jul. 2020

Publications

- [1] R. Alves, J. Cunha, and A. Madeira. “First Steps in a Paraconsistent Transition Systems Toolkit”. In: *Theoretical Aspects of Software Engineering - 20th International Symposium, TASE 2026, Shanghai, China, July 4-6, 2026, Proceedings*. Ed. by G. D. Giacomo, J. Li, and L. Xiao. Vol. 16697. Lecture Notes in Computer Science. Springer, 2026, pp. 368–385. DOI: 10.1007/978-3-032-30693-7_24.
- [2] J. Cunha, A. Madeira, and L. S. Barbosa. “Paraconsistent transition structures: compositional principles and a modal logic”. In: *Mathematical Structures in Computer Science* 35 (2025), e14. DOI: 10.1017/S0960129525100170.
- [3] J. Cunha, A. Madeira, and L. S. Barbosa. “Specification of paraconsistent transition systems, revisited”. In: *Science of Computer Programming* 240 (2025), pp. 103–196. DOI: 10.1016/J.SCIC0.2024.103196.
- [4] J. Cunha, A. Madeira, and L. S. Barbosa. “Paraconsistent Relations as a Variant of Kleene Algebras”. In: *Electronic Proceedings in Theoretical Computer Science* 421 (2025), pp. 130–147. ISSN: 2075-2180. DOI: 10.4204/eptcs.421.8.
- [5] J. Cunha, A. Madeira, and L. S. Barbosa. “Paraconsistent Reactive Graphs”. In: *Software Engineering and Formal Methods. SEFM 2024 Collocated Workshops - ReactS 2024 and CIFMA 2024, Aveiro, Portugal, November 4-5, 2024, Revised Selected Papers*. Ed. by J. Proença, R. Fervari, M. A. Martins, R. Kahle, and G. Pluck. Vol. 15551. Lecture Notes in Computer Science. Springer, 2024, pp. 105–111. DOI: 10.1007/978-3-031-94748-3_9.
- [6] J. Cunha, A. Madeira, and L. S. Barbosa. “Stepwise Development of Paraconsistent Processes”. In: *Theoretical Aspects of Software Engineering*. Ed. by C. David and M. Sun. Cham: Springer Nature Switzerland, 2023, pp. 327–343. ISBN: 978-3-031-35257-7.
- [7] J. Cunha, A. Madeira, and L. S. Barbosa. “Structured Specification of Paraconsistent Transition Systems”. In: *Fundamentals of Software Engineering - 10th International Conference, FSEN 2023, Tehran, Iran, May 4-5, 2023, Revised Selected Papers*. Ed. by H. Hojjat and E. Ábrahám.

Grants and Research Projects

- **Collaborator** in project **Banksy** | INESCITEC | AIBILI | CIDMA
Reference COMPETE2030-FEDER-00892000 2025 - 2029
- **PhD FCT Grant** | INESCITEC | CIDMA | Reference 2024.04652.BD 2025 - 2029
- **Collaborator** in project **Ibex** | INESCITEC | CISTER
Reference 10.54499/PTDC/CCI-COM/4280/2021 2023 - 2025
- **PhD PRR Grant** | CIDMA | Agenda ILLIANCE 2023 - 2024

Talks

- **20th Theoretical Aspects of Software Engineering Conference – (TASE26)**
Dishui Lake International Software Institute Jul. 6, 2026
China
- **Days in Logic 2026**
University of Minho Jan. 29, 2026
Portugal
- **4th Women in Mathematics Meeting in Portugal – (WM25)**
Faculty of Sciences of the University of Porto Jul. 16, 2025
Portugal
- **Quantitative Principles of Cyber-Physical Systems – (SEFM Workshop)**
University of Aveiro Nov. 4, 2024
Portugal
- **International Workshop on Reconfigurable Transition Systems – (ReacTs24)**
University of Aveiro Nov. 5, 2024
Portugal
- **19th International Workshop on Logical and Semantic Frameworks,
with Applications – (LSFA24), Universidade Federal de Goiás** Sep. 19, 2024
Brazil
- **Days in Logic 2024**
Técnico ULisboa Feb. 1, 2024
Portugal
- **17th Theoretical Aspects of Software Engineering Conference – (TASE23)**
University of Bristol Jul. 4, 2023
United Kingdom
- **10th IPM International Conference on. Fundamentals of Software
Engineering – (FSEN23), Institute for Research in Fundamental Sciences** May 4, 2023
Iran
- **World Logic Day in Aveiro**
University of Aveiro Jan. 17, 2023
Portugal

Invited Talks

- **CMUC Seminar of the Algebra, Logic and Topology Group**
University of Coimbra Dec. 16, 2025
Portugal

Teaching

- **Linear Algebra and Analytical Geometry**
BSc in Economics (9081) University of Aveiro
2025/2026
- **Linear Algebra and Analytical Geometry**
BSc in Computer and Informatics Engineering (8316) University of Aveiro
2024/2025
- **Linear Algebra and Analytical Geometry**
BSc in Informatics Engineering (9119) University of Aveiro
2023/2024

Academic Service

(External) Conference Reviewer: LSFA26 and NCL26.

Organizing Committees:

- Chair at the **5th Annual Meeting of DMat-UA PhD Students** Oct. 24, 2025
- Local Organizer of the **22nd International Conference on Software Engineering and Formal Methods – (SEFM24)** Nov. 4-8, 2024
- Chair at the **4th Annual Meeting of DMat-UA PhD Students** Oct. 25, 2024

Event Participation

TACL26 - Summer School, TASE26, XV Portuguese Category Seminar, Days in Logic 26, WM25, SEFM24, LSFA24, Days in Logic 24, TASE23, FSEN23

Industry Experience

Marketing Operations Delivery Jul. 2021 - Jul. 2022

Sonae MC | Porto, Portugal

- Responsible for documenting processes and training team members to execute communication campaigns in Salesforce Marketing Cloud.

Language

Portuguese (Native); **English** (Fluent); German (Beginner)